EFFECT OF CENTRAL CLEARING OF OTC DERIVATIVES

Learning Outcome

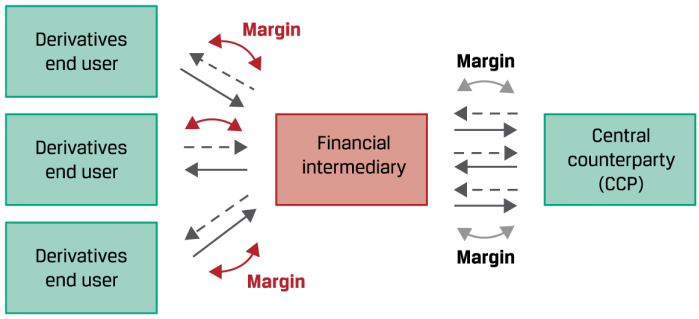
- explain why forward and futures prices differ

In periods of market and/or counterparty financial stress, large price movements combined with a derivative counterparty's inability to meet a margin call may force the closeout of a futures transaction prior to maturity. An OTC forward contract with more flexible credit terms, however, may remain outstanding.

The advent of derivatives central clearing, introduced in an earlier lesson, has created futures-like margining requirements for OTC derivative dealers who buy and sell forwards to derivatives end users. Dealers who are required to post cash or highly liquid securities to a central counterparty often impose similar requirements on derivatives end users. These dealer margin requirements reduce the difference in the cash flow impact of exchange-traded and OTC derivatives. This arrangement between dealers and their counterparties, shown in Exhibit 6, has been added to the original central clearing diagram from an earlier lesson.

Exhibit 6:

Margin Requirements for Centrally Cleared OTC Derivatives



▶ Expand / collapse Extended Description

Investors who actively use both exchange-traded futures or OTC forwards must therefore maintain sufficient cash or eligible collateral to fulfill margin or collateral requirements. Market participants must also consider the financing, transaction, and administrative costs of maintaining these positions when using derivatives in a portfolio.

Question Set: Forward and Futures Prices

1. Identify the following statement as true or false and justify your answer: If futures prices are positively correlated with interest rates, long futures contracts are more attractive than long forward positions for the same underlying and maturity.

Hide Solution

Solution:

rates.

2. An investor seeks to hedge its three-month MRR exposure on a £25,000,000 liability in two months and observes an implied forward rate today ($IFR_{2m,3m}$) of 2.95%. Calculate the settlement amounts if the investor enters a long pay-fixed (receive floating) FRA and a short futures contract, and compare and interpret the results if $MRR_{2m,3m}$ settles at 3.25%.

Hide Solution

Solution:

Solve for the pay-fixed FRA Cash Settlement (PV) value as follows:

Net Payment = (MRR_{B-A} – $IFR_{A,B-A}$) × Notional Principal × Period

= £18,750 (= [3.25% − 2.95%] × £25,000,000 × [1/4]).

The present value based upon $MRR_{2m,5m}$ of 3.25% is:

= £18,598.88 (= £18,750 / [1 + 0.0325/4]).

For the futures contract, contract BPV is equal to:

Contract BPV = £625 (= £25,000,000 × 0.01% × [1/4]).

For a 30-basis point increase in MRR (= 3.25% − 2.95%), the short futures contract will realize a price appreciation of £18,750 (= £625 × 30). Both contracts result in a gain from the investor's perspective as MRR rises. However, the futures settlement is larger due to the discounting of the FRA final payment to the settlement date.

3. Explain why short futures contracts are more attractive than short forward positions if futures prices are negatively correlated with interest rates for positions with the same underlying and maturity.

Hide Solution

Solution:

The reason that short futures contracts are more attractive than short forward positions if futures prices are negatively correlated with interest rates is because falling prices lead to futures profits that are reinvested in periods of rising interest rates, and rising prices lead to losses that occur in periods of falling interest rates.

4. Identify the following statement as true or false and justify your answer:

The convexity bias between interest rate futures and interest rate forwards causes the percentage price change to be greater (in absolute value) when MRR rises than when it falls for a forward than for a futures contract.

Hide Solution

Solution:

The statement is false. The convexity bias between interest rate futures and interest rate forwards causes the percentage price change to be greater (in absolute value) when MRR *falls* than when it *rises* for a forward contract, as opposed to a futures contract.

5. Explain how central clearing of derivatives reduces the difference in futures and forward prices for the same underlying and maturity periods.

Hide Solution

Solution:

derivatives end users. Dealers who are required to post cash or highly liquid securities to a central counterparty often impose similar requirements on derivatives end users. This arrangement between derivative dealers and their counterparties will reduce the difference in the cash flow impact of ETD and OTC derivatives. Hence, any price difference between ETD futures and OTC forwards will be reduced.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{CTM} CONTRACT TABLE MENU

The Bloomberg CTM function allows you to view the full category list of futures contracts and then click into each to see the commodity curve pricing by tenor.

> "CTM <GO> allows you to search for futures contracts, so you can easily locate commodities of interest based on category, exchange, or region." -Bloomberg

To explore this function, type CTM into the Bloomberg command line and press Enter or <GO>. Select one of the numbered menu items, such as Corn, to display a corresponding list of tickers. Then click on one of the tickers. Select CCRV from the popup menu to see the futures pricing curve.

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